

HANSEN NGAI

✉ ngaihl@mcmaster.ca
🌐 hansenngai.github.io/profile
📄 hansenngai

Skills

TECHNICAL

Java
Python
C
Object Oriented
Programming
HTML/CSS
Git
MATLAB

EMPLOYABILITY

Communication
Problem solving
Teamwork
Time Management
Multi-tasking
Adaptability

Awards

McMaster Honour
Award Level 2

Dean's Honours List

Outstanding
achievement in Grade
12 University Physics

Projects

Medieval War Simulation

- Applied principles of Object-Oriented Programming and Data Structures to create a simulation in Java
- Included path-tracking algorithm for AI
- Collaboration with other students to work efficiently

CD-ROM gear system

- Designed a gear system to implement inside a CD-ROM reader
- Worked with 3D modelling and simulation software
- Followed strict physical constraints
- Completed full engineering report

Experience

City of Markham

Robotics & Computers Camp Counselor

Markham, ON
Jun 2016 to Sep 2016

- Introduced concept of algorithms to solve problems
- Programmed with Scratch and Ozoblockly
- Created a positive and supporting environment with technology

Pierre Elliott Trudeau HS

VP Training for Computer Science Club

Sep 2014 to Jun 2015

- Led classes on principles of computer programming
- Taught topics such as Object-Oriented Programming, algorithms, data structures, Java syntax
- Coordinated lessons and workshops with other senior students

Camp Green Acres

Camp Counselor

Markham, ON
Jun 2015 to Jul 2015

- Maintained accurate records of events including attendance, incident reports, and logbook documentation
- Led daytime activities and followed strict schedule
- Communicated with parents about participant's experiences and raised concerns to head staff

Education

McMaster University

B.Eng Electrical Engineering (co-op) 2019

Hamilton, ON, Canada

Relevant courses: Engineering Computation, Principles of Programming, Data Structures & Algorithms, Logic Design

Activities

McMaster Formula Electric Racecar

Ontario Society of Professional Engineers

Student Mentorship Program (under PEO)

McMaster Engineering Society

Hackathon participant